

Cyberster Internship Program

Internship Report

TRACK: BLUE TEAM

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ROLL NUMBER: CSI-B1-121

WEEK 9: Advanced Browser Forensics, Shell/LNK Analysis, and Artifact Integration

1. Executive Summary

This report details the forensic triage and timeline reconstruction of a live Windows endpoint, focusing on user web activity and local file interactions. By extracting and correlating browser databases (Chrome/Edge) with Windows Shell artifacts (LNK files and Jump Lists), we reconstructed a unified chronological timeline of user actions.

Conclusion: The analyzed artifacts indicate structured, legitimate forensic and educational activity. The user downloaded forensic utilities, executed them locally, and accessed local development environments. No malicious extensions, unauthorized downloads, or suspicious file executions were detected.

2. Methodology & Evidence Integrity

Data acquisition prioritized evidence preservation. Browser processes were forcefully terminated prior to extraction to release active SQLite database locks and prevent data corruption.

- **Acquisition:** User profiles for Chrome and Edge were copied to a segregated C:\Forensics\BrowserData directory.

Chrome Profile:

```
C:\Windows\System32>robocopy "%localappdata%\Google\Chrome\User Data\Default" "C:\Forensics_Week9\Day60-
62_Browser\Chrome_Profile" /E

------------------------------------------------------------------------------
   ROBOCOPY     ::     Robust File Copy for Windows
------------------------------------------------------------------------------

  Started : Thursday, April 30, 2026 10:30:29 PM
   Source : C:\Users\mrymm\AppData\Local\Google\Chrome\User Data\Default\
     Dest : C:\Forensics_Week9\Day60-62_Browser\Chrome_Profile\

    Files : *.*

  Options : *.* /S /E /DCOPY:DA /COPY:DAT /R:1000000 /W:30
```

Edge Profile:

```
C:\Windows\System32>robocopy "%localappdata%\Microsoft\Edge\User Data\Default" "C:\Forensics_Week9\Day60
-62_Browser\Edge_Profile" /E

------------------------------------------------------------------------------
   ROBOCOPY     ::     Robust File Copy for Windows
------------------------------------------------------------------------------

  Started : Thursday, April 30, 2026 10:53:36 PM
   Source : C:\Users\mrymm\AppData\Local\Microsoft\Edge\User Data\Default\
     Dest : C:\Forensics_Week9\Day60-62_Browser\Edge_Profile\

    Files : *.*

  Options : *.* /S /E /DCOPY:DA /COPY:DAT /R:1000000 /W:30
```

- **Integrity Verification:** SHA-256 hashes were immediately generated for critical SQLite databases (History, Cookies, Web Data) to establish a defensible chain of custody.

```
PS C:\Windows\system32> Get-FileHash "C:\Forensics_Week9\Day60-62_Browser\Chrome_Profile\History" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          AA568E39509868111B29B85A1317AC2E8D2B12BE474C7363408A2E09D0736153       C:\Forensics_Week9\Day60-62_Browser\Chrome_Profile\History
```

```
PS C:\Windows\system32> Get-FileHash "C:\Forensics_Week9\Day60-62_Browser\Edge_Profile\History" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          308C5E6C4592A22B4372FD837F7A99E62BC4462AD7D99B92629F3F5673E2104A       C:\Forensics_Week9\Day60-62_Browser\Edge_Profile\History
```

3. Part A: Advanced Browser Forensics

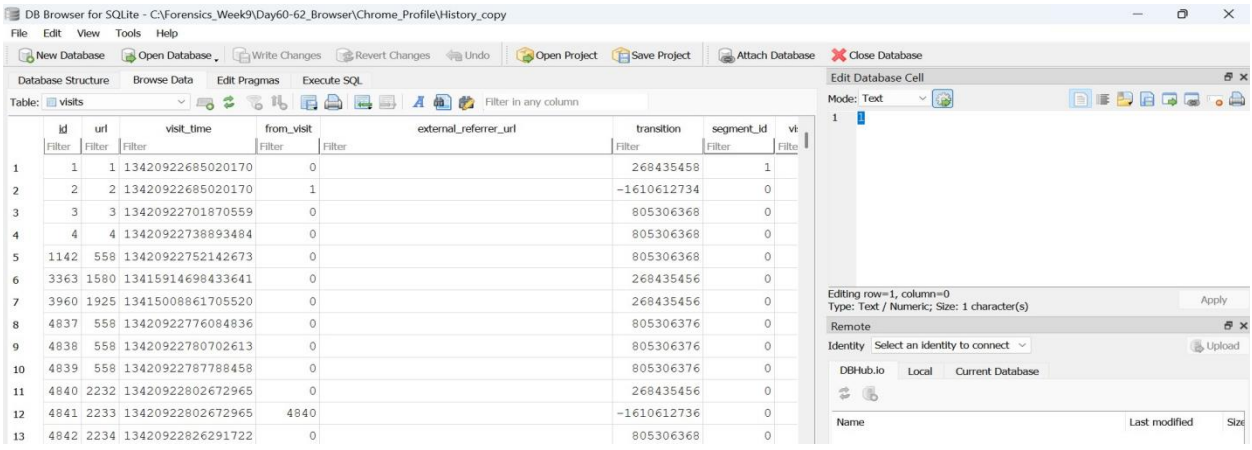
Browser artifacts were parsed using DB Browser for SQLite and NirSoft utilities to reconstruct web activity, downloads, and active sessions.

3.1. Web History & Download Activity

- **Browsing Patterns:** SQLite urls and visits tables confirmed traffic primarily directed toward educational resources, GitHub, and a local application (localhost - local phishing detection system).
- **Downloads:** The downloads table confirmed the deliberate acquisition of forensic and database tools. Identified files include:
  - LECmd.zip
  - JLECmd.zip
  - DB.Browser.for.SQLite-3.13.1-win32.zip

3.2. Cookies, Cache, & Extensions

- **Session Data:** Cookie analysis confirmed active, authenticated sessions corresponding to the observed web history.
- **Cache:** Extracted cache data revealed standard CDN assets and media associated with routine browsing.
- **Extensions:** An audit of installed browser extensions revealed only standard, benign add-ons. No unauthorized or malicious extensions were present.



4. Part B: Shell & LNK File Forensics

To track what happened *after* files were downloaded, we analyzed Windows Shell artifacts, which record local file and application interactions.

4.1. LNK (Shortcut) File Analysis

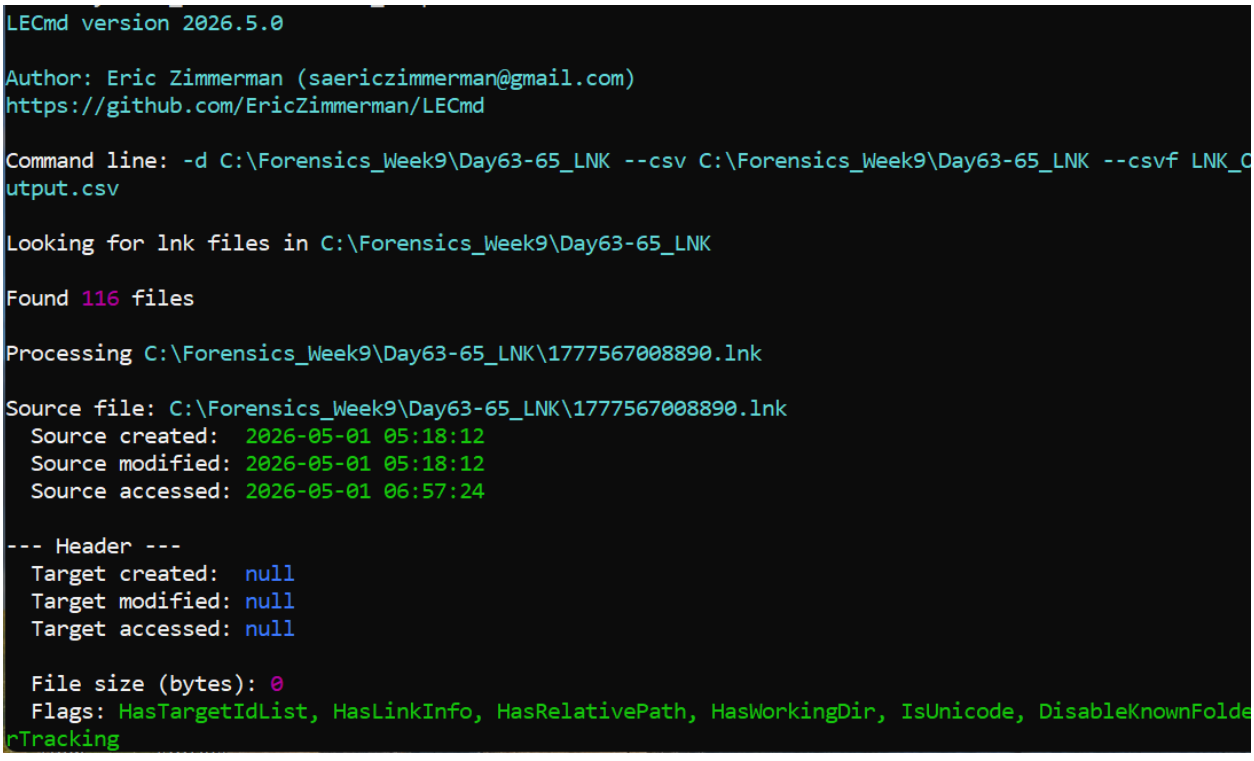
Using Zimmerman's LECmd, LNK files located in the Windows\Recent directory were parsed.

- **Findings:** The parsed output verified the execution of the tools downloaded in Part A. Specifically, LNK metadata confirmed the extraction and execution of LECmd.exe and DB Browser. Additional LNKs pointed to local documents, including CodeAlpha.pdf.

4.2. Jump List Analysis

Using JLECmd, AutomaticDestinations and CustomDestinations files were parsed to review taskbar and pinned-item interactions.

- **Findings:** Jump List data corroborated the LNK findings, showing sequential access to the downloaded ZIP archives and the local localhost development environment.



5. Part C: Timeline Generation & Artifact Integration

To provide a definitive sequence of events, browser timestamps (Webkit format) and Windows Shell timestamps (FILETIME) were normalized to UTC (PKT - 5 hours) and merged into a unified timeline.

5.1. Cross-Artifact Correlation

The integrated timeline successfully links external web activity to local system execution:

1. **Web Access:** User navigates to a repository (Browser History).
2. **File Acquisition:** User downloads LECmd.zip and DB Browser (Browser Downloads).
3. **Local Interaction:** User extracts the ZIP archives (LNK / Shell activity).
4. **Execution:** User launches the forensic tools and opens local PDF documents (LNK / Jump Lists).

Timestamp UTC	Artifact	Tool Used	Event Description	Original Path/URL	Analyst Note	Correlation	Significance
4/22/2026 4:32	Browser	BrowsingHistoryView	Accessed phishing detection system	http://127.0.0.1:5000	Local phishing tool used	Later file activity observed	Medium
4/28/2026 14:30	LNK	LECmd	Document opened	Banking-Law-1.pdf	File opened from Desktop	Normal usage	Low
5/1/2026 0:36	LNK	LECmd	Tool accessed	BrowsingHistoryView.chm	Forensic tool opened	Matches investigation activity	High
5/1/2026 0:38	LNK	LECmd	Folder accessed	browsinghistoryview	Tool folder opened	Related to tool usage	Medium
5/1/2026 1:23	LNK	LECmd	Tool file accessed	chromecacheview.zip	Another forensic tool used	Investigation activity	High
5/1/2026 1:57	LNK	LECmd	File accessed	cert.zip	Downloaded archive accessed	Possibly downloaded earlier	Medium
4/17/2026 15:07	LNK	LECmd	Document opened	D:\Internships\CodeAlpha.pdf	External file opened	Older activity	Low

6. Conclusion

The integration of Browser and Shell artifacts successfully validated the host's activity timeline. Correlating the origin of a file (Browser) with its local execution (LNK/Jump Lists) confirmed that the user's workflow was exclusively related to educational and forensic analysis tasks. The investigative framework utilized in Week 9 is highly effective for tracing the lifecycle of a file from web-download to local-execution in an incident response scenario.